



It's not all about Feeling Lucky

We dig deep into Web Search Engines, revealing their functions and methods.

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Before the advent of search engines, the web was a strange place. Back then, content was growing at a startling pace, with no means to filter information or direct people to content they were unaware of. Users typically used forums, IRC or email to find links. Any service that could make sense of the range of information, make it coherent, and allow users to find the content they wished for, was poised to grow spectacularly.

What was necessary, was a front door to the internet, a structure that allowed users to "surf". Within a decade of their introduction, search engines changed the entire face of the web. Before search engines, the web was mostly used for academic purposes and communication – primarily through electronic mail. Search engines allowed users to create content, market products, establish e-commerce and offer a wide variety of services. The technologies behind search engines have evolved with time, and are constantly mutating. The innovations, takeovers and competition among search engines were intense in the beginning of this century because the winner might replace Microsoft as the chief technology service provider. No guesses for who won, but the struggle is still on. There are a whole range of technologies driving a new generation and breed of search engines.

The internet is a connection of different servers and computers. There is a range of cable types that make these connections. These servers may belong to individuals, or be owned by a company. Stored on these servers are a number of documents written in a markup language (HTML and variants) that are interpreted by browsers. These documents invariably have hyperlinks (text that leads viewers to another page when clicked). This network of hyperlinks forms the world wide web, known simply as *the web*. While the internet is the hardware, the web is the information stored on this hardware. Search engines keep track of developments and changes on the web, and allow users to track down documents they want to access. This is a never ending process because the web is constantly changing with every passing moment.

The history of internet search

HotBot and AltaVista were among the early search engines. Both Yahoo! and MSN have had popular email and messenger services, both augmented by search. Yahoo! had the additional benefit of having a larger database of web sites stored in a directory structure. AskJeeves was also a very popular web service that was one of the first search engines to allow users to look for information as if they were talking to an actual person. AltaVista was one of the first search engines to have a wide audience, and growth in popularity because of the number of pages it had indexed. HotBot and Yahoo! dethroned AltaVista. While this race had been going on since 1996, Google entered the scene only in 1998 with a small index and minimalist interface and vision.

There is a whole range of highly specialised search engines developed in recent times, but essentially, the status quo has been maintained since 2000, with the top three search engines being Google, Yahoo! and MSN, in that order. Another important player is Inktomi that only provided back-end search technologies, and were used by both Yahoo! and MSN. Inktomi never had its own web portal for providing search services to users. They allowed their indexes and search technologies to be used by other search service providers. Before Google, Inktomi-based search providers were the most used, and displaced HotBot and AltaVista as the leading search service provider. It also had a large database of business and listed ads from these businesses next to the search results. Inktomi pioneered the pay-per-click model, which was later perfected by Google with its AdSense. Yahoo! switched to Google's databases and back to Inktomi within the five years of the millenium. MSN started using its own search technologies since 2004. Inktomi was acquired by Yahoo! in 2002.

With the web growing, search engines started increasing their services. Google was the market leader in innovation, buying out promising startups or stressing on developing fresh services in its own labs. AskJeeves.com was the first to make money by asking pages to pay for being indexed. Multimedia content provisions were made by early search engines such as Inktomi, but never really took off on a large scale. Search engines now make money by displaying advertisements related to a search query. In most engines, these ads occupy a separate space, but this distinction is not always clear. Multimedia search including video and images is offered by leading search providers. Locations, maps and people are other common search services offered by search engines, but are now feasible only in a few countries.

Spiders crawling the web

One of the fundamental tasks of a search engine is to find out that documents exist on the web at all. This was initially done by looking at known file servers, but the process is much more complicated now. Search engines use programs or scripts that automatically search the web, following hyperlinks. These programs or scripts are known as robots (colloquially called bots) or spiders. Spiders go to a particular web page, save it on their servers and follow all the links on that page. Going over the web is called "crawling", which is a continuous process. There are a number of approaches that these spiders can make when crawling the web. A wide crawl will focus on as many different URLs as possible, from as many different domains as possible. A depth crawl first goes deep into a single site. Practically, it is essentially a compromise between these approaches. There are also random crawls, which do not give any preference to a single approach. Spiders continuously track the creation of new content, and hence have a never-ending job. Search engine crawlers, typically, have a huge queue of URLs waiting to be crawled. Spiders will never be able to *crawl* the entire web, and no search engine has crawled more than 20 per cent of the web. The number of pages crawled by a spider is no measure of the quality of the search engine. Google